Problem Set 1
Econ 210 C

Juan Herreño

1 Instructions

This problem set is due April 26th. It is perfectly fine to work in groups, no matter the size of the group. It is ok to use AI tools, but I recommend you use them responsibly. You are responsible for the output that you present.

I try to give you sufficient context in the questions so that I believe it possible that a good AI agent can solve the problem set, but you know, that would not be any fun, and you'd be learning nothing, so by all means try to solve it, but if you want to experiment, ask AI for help, compare your code with that generated by AI, go for it, as long as you think this helps with your understanding. This pset is a chance for you to learn economics, take that chance.

You must submit your own solution, along with your codes. Please submit a .zip code with your whole material, and separately submit a .pdf with your solutions. Everything in your pdf must be generated by your codes. You must be able to explain everything you did and have a conversation about it.

I will choose a random number of students to come to office hours and have a conversation about their work with me.

Question 1. Time Aggregation

1.1. **Context:** Consider the quarterly New Keynesian Phillips Curve

$$\pi_t^q = \beta_q \mathbb{E}_t \pi_{t+1}^q + \kappa_q y_t^q, \tag{1}$$

where

$$\kappa_q = \frac{(1-\alpha_q)(1-\alpha_q\beta_q)}{\alpha_q}. \tag{2}$$

Define annual inflation as the sum of the four quarterly inflation rates in year τ:

$$\Pi_\tau = \pi_{4\tau}^q + \pi_{4\tau+1}^q + \pi_{4\tau+2}^q + \pi_{4\tau+3}^q. \tag{3}$$

Throughout, treat the quarterly model as the true data-generating process and use annual aggregation only as a measurement transformation.

1.2. **Quarterly Benchmark**

(a) Starting from the quarterly NKPC, derive the iterated-forward representation for π_t^q.

(b) In that representation, how are future output gaps weighted over time?

(c) What economic roles are played by β_q and κ_q?

1.3. **From Quarterly Inflation to Annual Inflation**

(a) Substitute the forward solution for each quarterly inflation term into the definition of Π_τ.

(b) After substitution, which quarterly output gaps affect annual inflation in year τ?

(c) Why does annual inflation place different effective weights on output gaps realized within year τ and output gaps realized after year τ?

1.4. **Within-Year Constant Output Gap**

Assume that the quarterly output gap is constant within each year, so that for some yearly object Y_τ,

$$y^q_{4\tau} = y^q_{4\tau+1} = y^q_{4\tau+2} = y^q_{4\tau+3} = Y_\tau.$$

(a) Under this restriction, how does the annual inflation expression simplify?

(b) Consider a one-year output-gap episode in which $Y_\tau \neq 0$ and $Y_{\tau+j} = 0$ for all $j \geq 1$. What is the effect of this episode on each of the four quarterly inflation rates in year τ?

(c) After summing across the four quarters, what does this imply about the coefficient linking annual inflation to Y_τ?

1.5. **Naive Annualization versus the Aggregated Result**

(a) Using the result from the previous question, write the annual inflation equation implied by the true quarterly model for the case of a one-year output-gap episode.

(b) Now suppose a researcher constructs annual Calvo and discounting parameters by setting

$$\alpha_a = \alpha_q^4, \qquad \beta_a = \beta_q^4, \qquad \kappa_a = \frac{(1-\alpha_a)(1-\alpha_a\beta_a)}{\alpha_a}.$$

and imposes those parameters into a NK Phillips curve naively. What annual Phillips curve would this researcher write down? Intuitively, give me a reason why you think this aggregation step may work and one reason you think it may not work?

(c) Compare the coefficient on Y_τ in the true aggregated equation to the coefficient implied by this annualization. What does the comparison suggest about mapping quarterly structural parameters into annual coefficients?

1.6. **Persistence and Annual Weights**

(a) Suppose output gaps can persist for more than one year. Why is the one-year calculation no longer sufficient?

(b) Write annual inflation in the form

$$\Pi_\tau = \kappa_q \sum_{m=0}^{\infty} w_m Y_{\tau+m}.$$

What determines the sequence $\{w_m\}_{m=0}^{\infty}$?

(c) Why is the contemporaneous weight w_0 different from the later weights w_m for $m \geq 1$?

1.7. **Failure of Form Invariance**

(a) Write the annual inflation equation one year ahead, for $\Pi_{\tau+1}$, and multiply it by β_q^4. Which terms line up with those in the equation for Π_τ?

(b) Which term does not line up exactly when you compare Π_τ with $\beta_q^4 \Pi_{\tau+1}$?

(c) Why does that mismatch imply that annual aggregation does not preserve the quarterly NKPC form?

Question 2. Monetary Policy Shocks, Local Projections, and Indirect Inference

For this question, use monthly U.S. data. Please use industrial production from FRED series `INDPRO`, the price level `CPIAUCSL`, the federal funds rate `FEDFUNDS`, unemployment `UNRATE`, and a commodity-price index `PPIACO`. For the monetary policy shock, use the monthly Romer-Romer shock file in Johannes Wieland's OpenICPSR archive, *Updated Romer-Romer Monetary Policy Shocks*:

`https://www.openicpsr.org/openicpsr/project/135741`

In that archive, the monthly file reports several shock measures; use `resid_full` as your baseline series.

2.1. **Cumulative Local Projections** Use the monthly sample 1969:01–2007:12 and horizons $h = 0, \ldots, 48$. Run the cumulative local projections directly. Define

$$\begin{aligned} Y_{t,h} &= \log(\texttt{INDPRO}_{t+h}) - \log(\texttt{INDPRO}_{t-1}), \\ P_{t,h} &= \log(\texttt{CPIAUCSL}_{t+h}) - \log(\texttt{CPIAUCSL}_{t-1}), \\ I_{t,h} &= \texttt{FEDFUNDS}_{t+h} - \texttt{FEDFUNDS}_{t-1}. \end{aligned}$$

Let

$$\Delta i_t = \texttt{FEDFUNDS}_t - \texttt{FEDFUNDS}_{t-1}.$$

For each variable $z \in \{Y, P, I\}$, estimate the IV local projection

$$z_{t,h} = \alpha_h^z + \beta_h^z \Delta i_t + \Gamma_h^{z\prime} W_t + u_{t+h}^z,$$

instrumenting Δi_t with $\texttt{resid_full}_t$. Use the control vector W_t consisting of the contemporaneous values of $\log(\texttt{INDPRO}_t)$, $\texttt{UNRATE}_t$, $\log(\texttt{CPIAUCSL}_t)$, and $\log(\texttt{PPIACO}_t)$, together with two lags of `resid_full`, `FEDFUNDS`, $\log(\texttt{INDPRO})$, `UNRATE`, $\log(\texttt{CPIAUCSL})$,

and log(`PPIACO`). Also estimate the reduced-form local projection

$$z_{t,h} = a_h^z + \psi_h^z \texttt{resid_full}_t + \Lambda_h^{z\prime} W_t + e_{t+h}^z.$$

Here the coefficient β_h^z is already the cumulative impulse response at horizon h. Normalize the IV responses so that $\beta_0^I = 1$, that is, a 100 basis point increase in the federal funds rate on impact. Report Newey-West standard errors with bandwidth $h+1$ and 95 percent confidence bands. Plot the cumulative IV impulse responses and report the reduced-form responses alongside them.

2.2. **Indirect Inference with $\sigma = 1$ and $\rho_i = 0$** Consider the New Keynesian model

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t \pi_{t+1}\right), \qquad \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t, \qquad i_t = \rho_i i_{t-1} + 1.5\pi_t + \varepsilon_t^m,$$

where the monetary policy shock follows

$$\varepsilon_t^m = \rho_m \varepsilon_{t-1}^m + u_t, \qquad \rho_m = 0.5.$$

With $\sigma = 1$ and $\beta = 0.99$, and $\rho_i = 0$, solve the model (perhaps using Dynare, gensys, or the toolbox of your preference) and compute the model-implied responses of the output gap, inflation, and the nominal interest rate to a one-time innovation u_t, normalized so that the impact response of i_t is 1. Convert model inflation into a model price-level response by cumulation:

$$\mathrm{IRF}_p^{\mathrm{model}}(h) = \sum_{j=0}^{h} \mathrm{IRF}_\pi^{\mathrm{model}}(j).$$

Then estimate (κ) by minimizing

$$\begin{aligned} Q(\kappa) = &\sum_{h=0}^{48} \left(\mathrm{IRF}_y^{\mathrm{cum}}(h) - \mathrm{IRF}_y^{\mathrm{model}}(h;\kappa)\right)^2 \\ &+ \sum_{h=0}^{48} \left(\mathrm{IRF}_p^{\mathrm{cum}}(h) - \mathrm{IRF}_p^{\mathrm{model}}(h;\kappa)\right)^2 \\ &+ \sum_{h=0}^{48} \left(\mathrm{IRF}_i^{\mathrm{cum}}(h) - \mathrm{IRF}_i^{\mathrm{model}}(h;\kappa)\right)^2. \end{aligned}$$

Report $(\widehat{\kappa})$ and overlay the model and empirical cumulative responses in one figure.

2.3. **Indirect Inference with $\sigma$and ρ_i Free** Repeat part (b), but now estimate (σ, κ, ρ_i)

jointly by minimizing

$$
\begin{aligned}
Q(\sigma,\kappa,\rho_i) = &\sum_{h=0}^{48}\left(\mathrm{IRF}_y^{\mathrm{cum}}(h) - \mathrm{IRF}_y^{\mathrm{model}}(h;\sigma,\kappa,\rho_i)\right)^2 \\
&+ \sum_{h=0}^{48}\left(\mathrm{IRF}_p^{\mathrm{cum}}(h) - \mathrm{IRF}_p^{\mathrm{model}}(h;\sigma,\kappa,\rho_i)\right)^2 \\
&+ \sum_{h=0}^{48}\left(\mathrm{IRF}_i^{\mathrm{cum}}(h) - \mathrm{IRF}_i^{\mathrm{model}}(h;\sigma,\kappa,\rho_i)\right)^2 .
\end{aligned}
$$

Report $(\widehat{\sigma},\widehat{\kappa},\widehat{\rho}_i)$, compare the restricted and unrestricted estimates, and discuss which features of the empirical impulse responses remain difficult for the model to match.

Question 3. Latent Natural Rate and VAR Approximation

Consider the standard New Keynesian model

$$
x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t\pi_{t+1} - r_t^n\right), \qquad \pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa x_t,
$$

$$
i_t = \rho_i i_{t-1} + \phi_\pi \pi_t + \varepsilon_t^m, \qquad r_t^n = \rho_n r_{t-1}^n + u_t^n, \qquad \varepsilon_t^m = \rho_m \varepsilon_{t-1}^m + u_t^m,
$$

where r_t^n denotes the natural interest rate and both exogenous disturbances follow AR(1) processes.

3.1. **State-Space Form and Blanchard-Kahn** Write the model as a first-order linear rational-expectations system. Some hints to make your life easier but that you can ignore: choose a state vector that includes the predetermined variables and the exogenous states. Write the model in matrix form. State the Blanchard-Kahn conditions and show that, when they hold, the equilibrium can be written as a linear state-space system with a VAR(1) transition.

3.2. **Eliminating the Natural-Rate from the observables**

Let the observable vector be

$$
z_t = \begin{pmatrix} x_t \\ \pi_t \\ i_t \end{pmatrix}.
$$

Suppose the econometrician does not observe r_t^n. Starting from the solved state-space system, eliminate r_t^n from the model and derive a reduced form in the observable vector z_t alone. Show that the resulting system is a VARMA process, state its order, and then rewrite it as a VAR. What is the order of that VAR?

3.3. **Estimating a Finite VAR(p)** Suppose an econometrician truncates the true observable law of motion and estimates a VAR(p) instead. What econometric and economic problems can arise from that approximation? What do you need to assume for this VAR to capture the true original system?